

DIESEL EXHAUST FLUID • ISO 22241

AdBlue & DEF Manufacturing

The 32.5% urea solution that lets a BS-VI diesel engine meet its
NO_x limit — blended, filtered and bottled in Bhandara, Maharashtra.

MANUFACTURER

PMR Industries
pmrindustries.com

01 THE PROBLEM

Nothing in diesel contains nitrogen. The air does.

Squeeze air hard enough and it passes 1600 °C. Above roughly 1300 °C the nitrogen and oxygen already in that air stop ignoring each other and combine. A diesel engine **manufactures its own pollutant** out of nothing but air and heat.

38,000

premature deaths worldwide in one year,
from diesels emitting more NO_x than their
standard allowed

78%

of the air entering every cylinder is
nitrogen, waiting for enough heat



DOWNSTREAM

REDUCED LUNG FUNCTION

GROUND-LEVEL OZONE

ACID RAIN

CROP YIELD LOSS

PARTICULATE HAZE

NITROGEN IN SOIL AND WATER

The limit collapsed. Demand did the opposite.

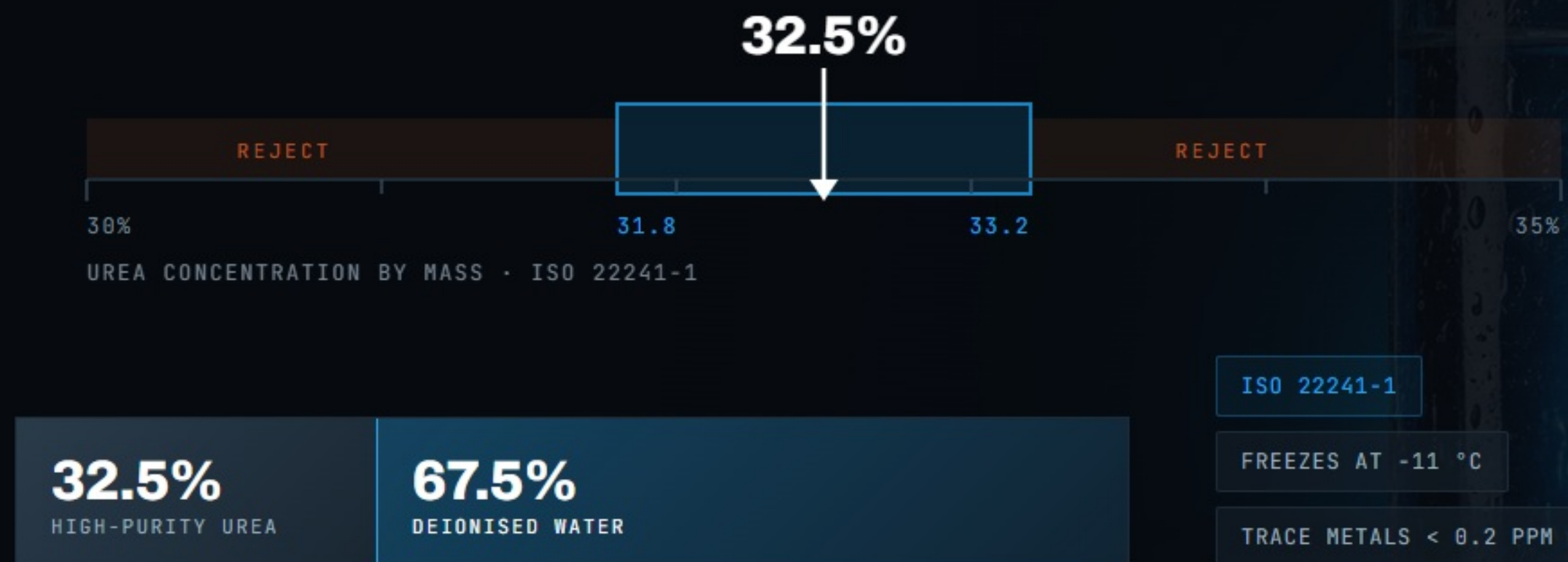
India skipped BS-V and went straight from BS-IV to BS-VI in April 2020. No engine meets that number by burning cleaner alone — it meets it with an SCR catalyst, and a catalyst is inert without DEF.



Sources: BS-VI vs BS-IV heavy-duty NO_x reduction, Ministry of Environment. Market values, BlueWeave Consulting, India Diesel Exhaust Fluid (AdBlue) Market 2024-2030.

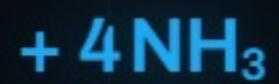
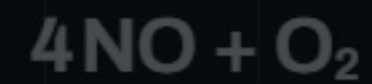
The recipe is easy. The window is not.

DEF is urea and water. Anyone can mix that. What it has to be is **32.5% urea, every single batch** — outside a band 1.4 points wide the catalyst stops converting, or starts failing. That band is the whole business.

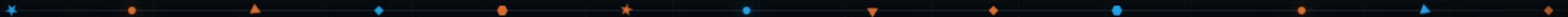


DEF never enters the engine.

It is metered into the hot exhaust downstream. Heat splits the urea into ammonia; the ammonia meets NO_x on the catalyst surface.



UREA DECOMPOSES TO AMMONIA



We build the fluid, end to end.

CONFORMS TO
**ISO
22241-1**

- UREA 31.8 - 33.2%
- REFRACTIVE INDEX
- ALKALINITY AS NH_3
- TRACE METALS < 0.2 PPM



UREA CONCENTRATION

32.5%

04

Lab testing

Every batch is checked on refractive index, conductivity and alkalinity before it can leave the tank.

ISO 22241-1

01
Water purification

02
Urea blending

03
Filtration

04
Lab testing

05
Storage

06
Filling & capping

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